

LOW REN HWA

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Work Experience

AI Engineer – Carro

August 2025 – Present

- Led development of Carro's customer service agent fleet, partnering with stakeholders to translate business needs into 12 production agentic workflows across Singapore, Malaysia, and Hong Kong, servicing 60,000+ tickets monthly.
- Built custom agent harnesses to automate internal data retrieval, reporting, and operational workflows across departments.
- Reduced LLM costs by 80% through agent harness optimization and intelligent model routing.
- Built Carro's core AI infrastructure on GCP, deploying *LiteLLM* for model routing and *Langfuse* for tracing and evaluation.
- Developed *n8n* automations for customer outreach, driving 30% revenue growth for Singapore's workshops.

AI Engineer – AI Singapore

June 2024 – April 2025

- Delivered a GenAI MVP for a gaming industry project, generating 3D character animations from text.
- Built end-to-end ML pipelines to train and deploy state-of-the-art diffusion models using *PyTorch*.
- Used multimodal video LLMs for automated data labelling, improving text-to-motion metrics by 20% over baseline.
- Optimized *Gitlab* CI/CD pipelines, resulting in an 80% reduction in testing and deployment time.

Research Assistant – National Institute of Education

October 2023 – March 2024

- Developed a speaker diarization system with professors to measure student participation and learning outcomes.
- Implemented an automatic speech recognition (ASR) and diarization pipeline with Nvidia's *Nemo*, fine tuning models with self-annotated transcripts to outperform open-source benchmarks by 30%.

Machine Learning Engineer Intern – Singapore Department of Statistics

May 2023 – August 2023

- Partnered with various departments to prototype NLP solutions for text classification, Q&A, and automated report generation using *LangChain* and GPT-3.5.

Database Administrator Intern – Tesseract.io

July 2022 – February 2023

- Maintained production databases with *PL/pgSQL* and *Python scripts*, migrating 300,000+ records for 10+ law firms.

Education

B.E. Computer Science, Nanyang Technological University, Singapore

Aug 2020 – June 2024

Personal Projects

Kanta: Smart collaborative film camera app for events

April 2025 – May 2025

[GitHub](#) | Basically google drive but for sharing pictures, with inbuilt face detection and smart photo organization.

- Designed with a *FastAPI* backend, using Azure Blob Storage and Azure PostgreSQL to support fast image retrieval and queryability, and deployed using Azure Container Apps. Unsupervised clustering job executed with AWS Lambda.

Echo: Personalized LLM Digital Clone

April 2025 – May 2025

[GitHub](#) | Built a digital clone of myself using 5 years' worth of personal telegram data.

- Fine-tuned open-source LLMs (LLaMA 3, Gemma 3) using *Unsloth* for fast memory-efficient training. Built a modular microservice architecture with a *FastAPI* backend, and used AWS S3 for scalable data storage of telegram data.

Technologies & Languages

- **Languages:** Python, TypeScript, SQL
- **Technologies:** Git, Docker, Bash Scripting, FastAPI, Django, n8n
- **AI & Agents:** Langchain, Langgraph, Google ADK, Hermes, Nanobot, Langfuse, LiteLLM
- **Cloud & Infra:** Terraform, GCP (Compute Engine, Cloud Run, BigQuery), Azure (Blob Storage, Container Apps)

Interests: Rock climbing, Cooking, Guitar